

A Counterexample to the Weak-to-Strong Hardy Norm Law Below $p = 2$

arXiv automatic math run `fa_banach_001`

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Status

This packet gives a likely valid negative answer to Question 19(a) of Laitila–Tylli [1]. The question asks whether the norm estimate in Theorem 18(2), valid for $2 < p < \infty$, extends to $1 \leq p < 2$. It does not: radial composition symbols on $X = \ell_2$ have norm of order $\|C_\varphi\|_{HS}$, not $\|C_\varphi\|_{HS}^{2/p}$.

Source Question

The source is J. Laitila and H.-O. Tylli, *Composition operators on vector-valued analytic function spaces: a survey*, arXiv:1505.01945 [1]. The relevant statement appears as Question 19(a) on PDF page 17, after Theorem 18.

Theorem 18. *Let X be any infinite-dimensional complex Banach space.*

- (1) *If $\|C_\varphi\|_{HS} < \infty$, then C_φ is bounded $wH^p(X) \rightarrow H^p(X)$ for any p satisfying $1 \leq p < \infty$.*
- (2) *The norm $\|C_\varphi: wH^p(X) \rightarrow H^p(X)\|$ is equivalent to $\|C_\varphi\|_{HS}^{2/p}$ for $2 < p < \infty$.*
- (3) *$\|C_\varphi: wH^2(X) \rightarrow H^2(X)\| = \|C_\varphi\|_{HS}$.*

Parts (2) and (3) are obtained by explicit computations for $X = \ell^2$. The extension to arbitrary Banach spaces X is based on Dvoretzky's ℓ_2^n -theorem and coefficient multiplier results [14] corresponding to bounded operators

$$\sum_k a_k z^k \mapsto \sum_k \lambda_k a_k z^k, \quad H^2 \rightarrow H^p,$$

where (λ_k) is a fixed sequence. By contrast to Theorem 18, $\widetilde{C}_\varphi$ is bounded $wBMOA(\ell^2) \rightarrow BMOA(\ell^2)$ if and only if the scalar-valued operator C_φ is bounded $\mathcal{B} \rightarrow BMOA$, see [30, Example 4.1], where $\mathcal{B}$ is the Bloch space.

Question 19. (a) Does part (2) of Theorem 18 extend to $1 \leq p < 2$? The corresponding coefficient multiplier theorems $H^2 \rightarrow H^p$ for $1 \leq p < 2$ are not readily useful.

(b) Characterize the weakly compact compositions $\widetilde{C}_\varphi$ from $wH^1(X)$ to $H^1(X)$ if X is reflexive. It is possible to show that if

$$\lim_{s \rightarrow 1} \sup_{0 < r < 1} \int_{|\varphi(r\zeta)| > s} \frac{1}{1 - |\varphi(\zeta)|^2} dm(\zeta) = 0,$$

then $\widetilde{C}_\varphi$ is weakly compact from $wH^1(X)$ to $H^1(X)$ (details omitted). Note that $\widetilde{C}_\varphi$ is never compact $wH^1(X) \rightarrow H^1(X)$ for infinite-dimensional X by section 2.

In the notation of [1], $H^p(X)$ is the strong vector-valued Hardy space and $wH^p(X)$ is the weak Hardy space with norm

$$\|f\|_{wH^p(X)} = \sup_{\|x^*\| \leq 1} \|x^* \circ f\|_{H^p}.$$

Question 19(a) asks whether

$$\|C_\varphi: wH^p(X) \rightarrow H^p(X)\| \asymp \|C_\varphi\|_{HS}^{2/p}$$

continues to hold for $1 \leq p < 2$.

Proof Intuition

For $p \leq 2$, the weak H^p norm controls each Taylor coefficient of an ℓ_2 -valued function separately. After radial composition by $\varphi_r(z) = rz$, the strong $H^p(\ell_2)$ norm is bounded by the strong $H^2(\ell_2)$ norm, and Parseval reduces this to the square sum $\sum r^{2n} \|x_n\|^2$. This gives the upper bound $(\sum r^{2n})^{1/2}$. The test function

$$F(z) = (1, z, z^2, \dots)$$

has weak $H^p(\ell_2)$ -norm 1, and after radial composition its ℓ_2 -norm on the boundary is exactly $(\sum r^{2n})^{1/2}$. Thus the operator norm is exactly the Hilbert–Schmidt norm, which grows too slowly to be comparable with $\|C_{\varphi_r}\|_{HS}^{2/p}$ when $p < 2$.

Theorem

Theorem 1. *Let $1 \leq p \leq 2$, let $X = \ell_2$, and let $\varphi_r(z) = rz$ for $0 < r < 1$. Then*

$$\|C_{\varphi_r} : wH^p(\ell_2) \rightarrow H^p(\ell_2)\| = \|C_{\varphi_r} : H^2 \rightarrow H^2\|_{HS} = (1 - r^2)^{-1/2}.$$

Consequently the estimate

$$\|C_\varphi : wH^p(X) \rightarrow H^p(X)\| \asymp \|C_\varphi\|_{HS}^{2/p}$$

from Theorem 18(2) of [1] cannot extend to any $1 \leq p < 2$.

Proof. Write an arbitrary $f \in wH^p(\ell_2)$ as

$$f(z) = \sum_{n=0}^{\infty} x_n z^n, \quad x_n \in \ell_2.$$

For every unit vector $y \in \ell_2$, the scalar function

$$\langle f, y \rangle(z) = \sum_{n=0}^{\infty} \langle x_n, y \rangle z^n$$

belongs to H^p and has norm at most $\|f\|_{wH^p(\ell_2)}$. The n -th Taylor coefficient functional on H^p has norm at most 1, since

$$|a_n| = \left| \int_{\mathbb{T}} g(\zeta) \bar{\zeta}^n dm(\zeta) \right| \leq \|g\|_{L^1(\mathbb{T})} \leq \|g\|_{H^p}.$$

Therefore

$$\|x_n\|_{\ell_2} = \sup_{\|y\|_2=1} |\langle x_n, y \rangle| \leq \|f\|_{wH^p(\ell_2)}$$

for every n .

Since $1 \leq p \leq 2$, the normalized measure on $\mathbb{T}$ gives $\|h\|_{L^p(\mathbb{T}; \ell_2)} \leq \|h\|_{L^2(\mathbb{T}; \ell_2)}$. Thus Parseval yields

$$\begin{aligned} \|C_{\varphi_r} f\|_{H^p(\ell_2)} &\leq \|C_{\varphi_r} f\|_{H^2(\ell_2)} \\ &= \left(\sum_{n=0}^{\infty} r^{2n} \|x_n\|_{\ell_2}^2 \right)^{1/2} \\ &\leq \left(\sum_{n=0}^{\infty} r^{2n} \right)^{1/2} \|f\|_{wH^p(\ell_2)}. \end{aligned}$$

Hence

$$\|C_{\varphi_r} : wH^p(\ell_2) \rightarrow H^p(\ell_2)\| \leq (1 - r^2)^{-1/2}.$$

For the reverse inequality, define

$$F(z) = (1, z, z^2, \dots) \in \ell_2 \quad (z \in \mathbb{D}).$$

For any $y = (y_n) \in \ell_2$ with $\|y\|_2 \leq 1$,

$$\langle F, y \rangle(z) = \sum_{n=0}^{\infty} \overline{y_n} z^n$$

has H^2 -norm $\|y\|_2 \leq 1$, and hence H^p -norm at most 1. Taking $y = e_0$ shows $\|F\|_{wH^p(\ell_2)} = 1$. On the boundary,

$$\|F(r\zeta)\|_{\ell_2} = \left(\sum_{n=0}^{\infty} r^{2n} \right)^{1/2} = (1 - r^2)^{-1/2},$$

independently of $\zeta \in \mathbb{T}$. Therefore

$$\|C_{\varphi_r} F\|_{H^p(\ell_2)} = (1 - r^2)^{-1/2},$$

which proves equality.

Finally, on scalar H^2 , C_{φ_r} is diagonal with respect to the orthonormal monomial basis z^n :

$$C_{\varphi_r}(z^n) = r^n z^n.$$

Thus

$$\|C_{\varphi_r} : H^2 \rightarrow H^2\|_{HS}^2 = \sum_{n=0}^{\infty} r^{2n} = (1 - r^2)^{-1}.$$

For $1 \leq p < 2$, the quotient

$$\frac{\|C_{\varphi_r} : wH^p(\ell_2) \rightarrow H^p(\ell_2)\|}{\|C_{\varphi_r} : H^2 \rightarrow H^2\|_{HS}^{2/p}} = \|C_{\varphi_r}\|_{HS}^{1-2/p}$$

tends to 0 as $r \uparrow 1$. Hence no lower equivalence constant can hold for the proposed extension of Theorem 18(2) to $1 \leq p < 2$. $\square$

Verification Notes

The proof uses only the elementary coefficient estimate $|\widehat{g}(n)| \leq \|g\|_{H^p}$, the monotonicity $\|h\|_{L^p} \leq \|h\|_{L^2}$ on a probability space for $p \leq 2$, and Parseval in $H^2(\ell_2)$. The example is within the hypotheses of the source question because ℓ_2 is an infinite-dimensional complex Banach space and $\varphi_r(z) = rz$ is an analytic self-map of $\mathbb{D}$ with Hilbert–Schmidt scalar composition operator on H^2 .

The result disproves only the norm-equivalence extension asked in Question 19(a). It does not challenge Theorem 18(1), since the radial symbols still give bounded maps $wH^p(\ell_2) \rightarrow H^p(\ell_2)$.

Novelty Check

Before promotion, local run indexes were searched for the source arXiv id, weak Hardy spaces, weak-to-strong composition operators, Hilbert–Schmidt composition operators, and radial-dilation variants; no duplicate packet or attempt was found. Web searches on 2026-06-14 for exact phrases around $wH^p(X)$, $H^p(X)$, Hilbert–Schmidt composition operators, $1 \leq p < 2$, radial dilation, and the paper title found the source paper but no later exact answer. Because the argument is short, human review should still check whether this observation appears in non-arXiv literature.

Human Review Recommendation

The key review point is interpretive: confirm that “extends” in Question 19(a) means a uniform two-sided equivalence with constants independent of the symbol φ , as in Theorem 18(2). Under that standard reading, the radial family $\varphi_r(z) = rz$ gives a direct negative answer.

References

- [1] J. Laitila and H.-O. Tylli, *Composition operators on vector-valued analytic function spaces: a survey*, arXiv:1505.01945, 2015.